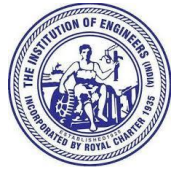




ST. VINCENT PALLOTTI
COLLEGE OF ENGINEERING & TECHNOLOGY, NAGPUR
AN AUTONOMOUS INSTITUTION



TATA
CONSULTANCY
SERVICES



International Conference On

AI Driven Engineering & Technology

AIDCon-2025

DECEMBER 12-13 , 2025

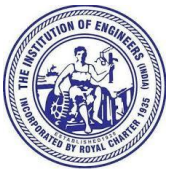




ST. VINCENT PALLOTTI
COLLEGE OF ENGINEERING & TECHNOLOGY, NAGPUR
AN AUTONOMOUS INSTITUTION



TATA
CONSULTANCY
SERVICES



Track: Healthcare Technologies & Systems

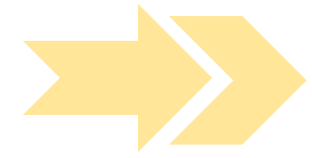
Paper ID: 81

Paper Title: A Review on Recent Advancements in Scoliosis Detection System

Author Name: Devanshu Ekhar

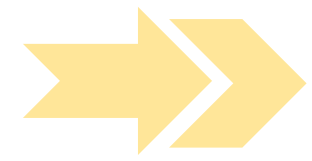
Project Guide: Dr. Yogesh Golhar

Co-author(s) Name: Jeena Joseph, Siddhant Lohakar, Saveri Dongre



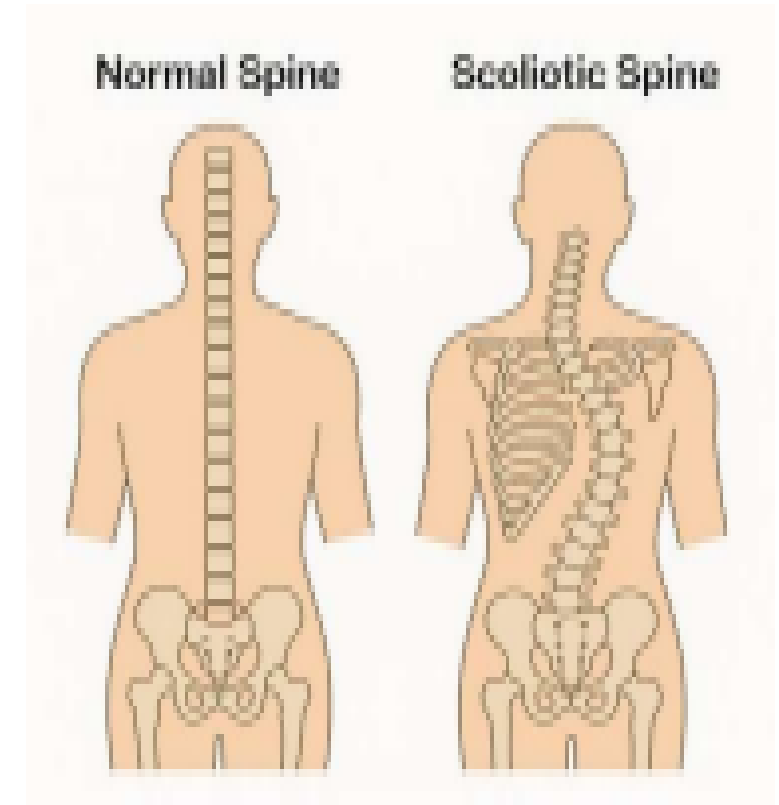
Outline of the Presentation

- 1. INTRODUCTION**
- 2. METHODOLOGY**
- 3. RESULTS AND DISCUSSIONS**
- 4. CONCLUSION**
- 5. REFERENCES**



Introduction

- Scoliosis = 3D spinal deformity affecting millions of adolescents worldwide.
- Traditional diagnosis relies on **X-ray + Cobb angle** → radiation + subjective variability.
- Modern research explores **AI, 3D imaging, genetics, MRI-based early detection**.
- Reviewed **30–35 high-impact papers (1996–2024)** across four domains.

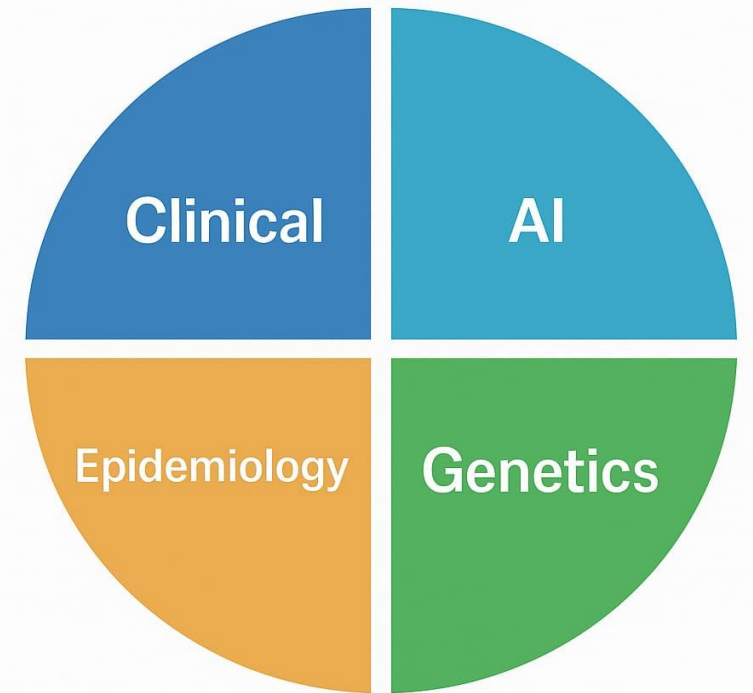


Comparison between a normal spine



Aim of the Review

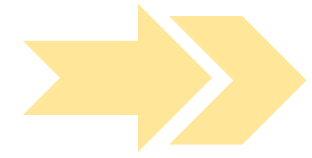
- Summarize advancements across **clinical, AI/deep learning, genetics, and epidemiology**.
- Identify **research gaps** in radiation-free screening & early prediction.
- Support development of **non-invasive, automated, school-level screening tools**.





Methodology

- **Databases:** PubMed, IEEE Xplore, Springer, Google Scholar.
- **Keywords:** AIS detection, X-ray analysis, CNN scoliosis, genetic markers, MRI scoliosis.
- **Screening:**
 - 135 articles found
 - 52 screened
 - **35 included** (relevant + high quality).
- **Evaluation based on:**
 - Input modality
 - Method/Model
 - Accuracy & clinical feasibility
 - Limitations

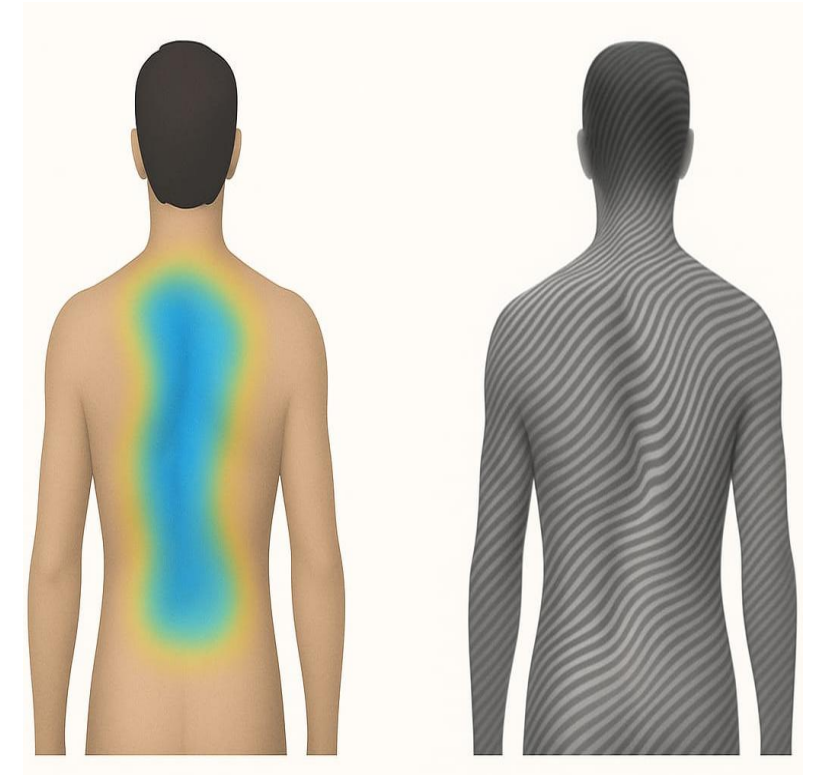


Key Clinical Insights

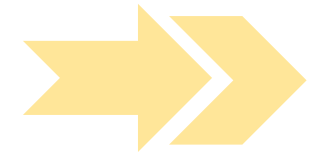
- MRI detects hidden spinal abnormalities often missed in plain radiographs.
- Early detection + bracing reduces curve progression significantly.
- Most AIS back pain is benign → helps reduce overuse of imaging.
- Surgical thresholds standardized ($\sim 45\text{--}50^\circ$ Cobb angle).

Imaging Advances (Surface Topography & 3D Systems)

- Surface topography reduces radiation exposure.
- Moiré topography used in school screening (Japan, Korea).
- 3D body scanning improves curve estimation accuracy.
- Several studies show **80–90% correlation with Cobb angle**.



3D depth map and Moiré pattern for scoliosis surface imaging.



AI & Deep Learning Models

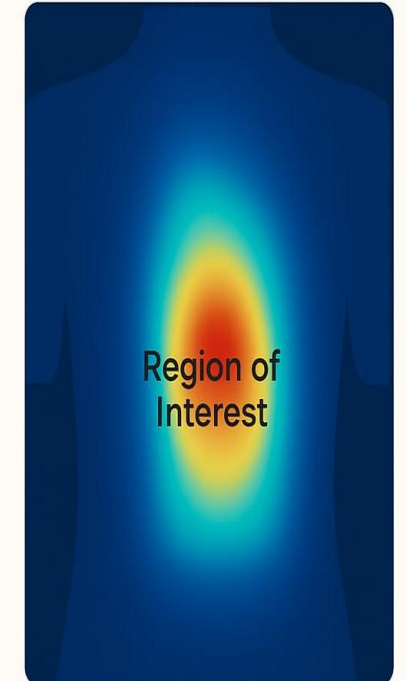
- CNNs trained on back photographs achieve **AUC 0.90–0.97**.
- DenseNet, ResNet, EfficientNet widely used for severity classification.
- Capsule Networks predict scoliosis progression better than CNNs.
- Depth-sensor + neural networks: $r = \mathbf{0.90+}$ correlation with Cobb angle.
- AI reduces radiation exposure & enables school screenings.

AI-Based Cobb Angle Automation

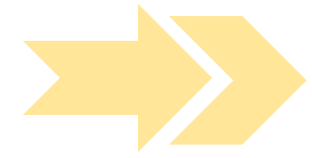
- Automated key-point detection improves accuracy.
- Deep learning achieves $<3^\circ$ error margin in many studies.
- Helps radiologists reduce time + improve consistency.
- Multi-view X-ray systems improve robustness.



**VERTEBRA
LANDMARKS**

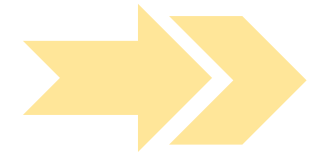


**HEATMAP OUTPUT
FROM A CNN**



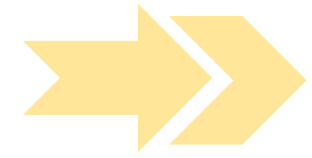
Mobile/Low-Cost Screening Technologies

- Smartphone apps using posture images for preliminary diagnosis.
- ML models usable in rural/low-resource settings.
- Several papers show >85% screening accuracy.



Genetic Findings

- Strong association with LBX1, GPR126, CHD7, and SOX9-related pathways.
- Possible X-linked inheritance in familial scoliosis.
- Genetic markers may help identify children at higher risk.



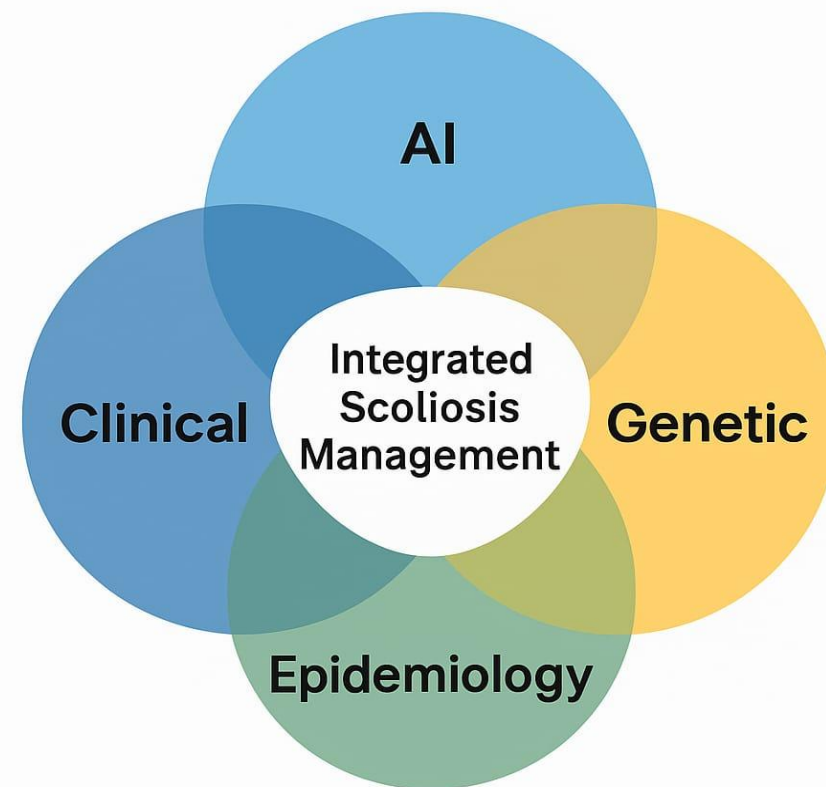
Epidemiological Insights

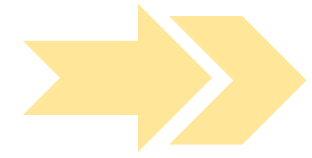
- Prevalence 2–4% globally.
- Majority of scoliosis cases are mild; <1% require surgery.
- Strong regional variation in detection → need for screening programs.
- Girls are 7–10x more likely to progress.



Results & Discussions

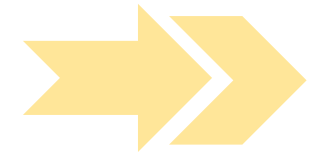
- **Early detection** dramatically improves outcomes.
- AI + 3D imaging = future of **radiation-free screening**.
- Genetics supports **risk prediction**, not diagnosis (yet).
- National screening programs reduce surgical burden.





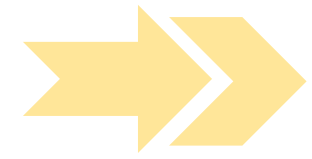
Limitations Identified Across Studies

- Lack of large, multi-country datasets.
- Deep learning models not externally validated in most papers.
- Limited research on **Indian adolescent population**.
- Genetic studies not standardized across labs.



Conclusions

- Significant progress made across **clinical, AI, imaging, and genetics**.
- AI-based tools show great promise for **school-based mass screening**.
- Future scoliosis diagnosis will be **non-invasive, automated, and data-driven**.
- Review of 35 papers confirms strong direction toward **precision medicine**.



Future Scope

- Develop multi-modal AI combining images + posture + genetics.
- Large-scale Indian dataset for scoliosis screening.
- Mobile app + depth sensor for real-time school screening.
- Nationwide, radiation-free screening program.



References

- [1] Evans, S. C., Edgar, M. A., Hall-Craggs, M. A., et al., “MRI of ‘idiopathic’ juvenile scoliosis: A prospective study,” *J. Bone Joint Surg. Br.*, 78(2), 314–317, 1996.
- [2] Ramirez, N., Johnston, C. E., Browne, R. H., “The prevalence of back pain in children who have idiopathic scoliosis,” *J. Bone Joint Surg. Am.*, 79(3), 364–368, 1997.
- [3] Justice, C. M., Miller, J. N., Marosy, B., et al., “Familial idiopathic scoliosis: Evidence of an X-linked susceptibility locus,” *Spine*, 28(6), 589–594, 2003.
- [4] Khanna, G., “Role of imaging in scoliosis,” *Pediatr. Radiol.*, 39(S2), S247–S251, 2009.
- [5] Mo, F., Cunningham, M. E., “Pediatric scoliosis,” *Curr. Rev. Musculoskelet. Med.*, 4(2), 91–97, 2011.
- [6] Fan, Y.-H., Song, Y.-Q., Chan, D., et al., “SNP rs11190870 near LBX1 is associated with adolescent idiopathic scoliosis,” *J. Hum. Genet.*, 57(4), 244–246, 2012.
- [7] Gervais, J., Périé, D., Parent, S., et al., “MRI signal distribution within the intervertebral disc as a biomarker of scoliosis,” *BMC Musculoskelet. Disord.*, 13, 239, 2012.
- [8] Konieczny, M. R., Senyurt, H., Krauspe, R., “Epidemiology of adolescent idiopathic scoliosis,” *J. Child Orthop.*, 7(1), 3–9, 2013.
- [9] Londono, D., Kou, I., Johnson, T. A., et al., “Meta-analysis identifies AIS association with LBX1 locus,” *J. Med. Genet.*, 51(6), 401–406, 2014.
- [10] Thérourx, J., Le May, S., Fortin, C., et al., “Back pain in AIS patients: Retrospective study,” *Pain Res. Manag.*, 20(3), 153–157, 2015.
- [11] Chettier, R., Nelson, L., Ogilvie, J. W., et al., “Haplotypes at LBX1 have distinct inheritance patterns in AIS,” *PLoS One*, 10(2), e0117708, 2015.
- [12] Grauers, A., Wang, J., Einarsdottir, A. Y., et al., “LBX1 as a susceptibility gene for idiopathic scoliosis,” *Spine J.*, 15(10), 2239–2246, 2015.



References Continue

- [13] Balagué, F., Pellisé, F., “Adolescent idiopathic scoliosis and back pain,” *Scoliosis Spinal Disord.*, 11, 27, 2016.
- [14] Wong, A. Y. L., Cheung, S. H., Cheung, J. P. Y., et al., “Back pain and biopsychosocial factors in AIS,” *Clin. Orthop. Relat. Res.*, 477(4), 676–691, 2019.
- [15] Yang, J. H., Cheung, W. C., Cheung, J. P. Y., et al., “Deep learning algorithms for scoliosis screening using back images,” *Commun. Biol.*, 2, 390, 2019.
- [16] Mohamed, M. A. E., Trivedi, J. M., Davidson, N., Munigangaiah, S., “A review of current concepts in AIS,” *Orthop. Trauma.*, 34(6), 338–345, 2020.
- [17] Luo, M., Zhang, Y., Huang, S., Song, Y., “Functions of LBX1 gene in AIS,” *Front. Genet.*, 11, 614984, 2021.
- [18] Sung, S., Chae, H.-W., Lee, H. S., et al., “Incidence and surgery rate of idiopathic scoliosis,” *Int. J. Environ. Res. Public Health*, 18(15), 8152, 2021.
- [19] Wang, H., Zhang, T., Cheung, K. M. C., et al., “Deep learning to predict progression in AIS,” *EClinicalMedicine*, 42, 101220, 2021.
- [20] Thomas, J. J., McCulloch, L. R., Sanders, J. L., et al., “Trends in incidence of AIS,” *J. Pediatr. Orthop.*, 41(6), 327–332, 2021.
- [21] Lee, G. B., Priefer, D. T., Priefer, R., “Scoliosis: Causes and treatments,” *Adolescents*, 2(2), 220–234, 2022.
- [22] Fraiwan, M., Audat, Z., Fraiwan, L., Manasreh, T., “Deep transfer learning for scoliosis detection,” *PLoS One*, 17(5), e0267851, 2022.
- [23] Hong, D., Lee, J. H., Kwon, J. W., “Scoliosis-related medical services in South Korea,” *Spine*, 47(11), 727–735, 2022.
- [24] Ishikawa, Y., Kokabu, T., Yamada, K., et al., “Predicting Cobb angle using DL + depth sensor data,” *J. Clin. Med.*, 12(2), 499, 2023.



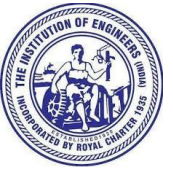
References Continue

- [25] Chu, K., Kuang, X., Cheung, P. W. H., et al., “Integrating 2D imaging and clinical info for AIS prediction,” *Global Spine J.*, 2023.
- [26] Li, M., Nie, Q., Liu, J., Jiang, Z., “Prevalence of scoliosis: Systematic review,” *Front. Pediatr.*, 12, 1399049, 2024.
- [27] Li, L., Wong, M.-S., “Machine learning for AIS progression prediction: Review,” *Biomed. Eng. Online*, 23, 80, 2024.
- [28] Zhang, L., Pei, B., Zhang, S., et al., “New scoliosis screening method using deep learning,” *Global Spine J.*, 2024.
- [29] Wang, S., Li, M., Ren, J., et al., “Global prevalence & associated risk factors of scoliosis,” *BMC Public Health*, 25, 3640, 2025.
- [30] Jinnah, A. H., Dolan, L. A., Weinstein, S. L., “AIS: Advances in diagnosis and treatment,” *Curr. Rev. Musculoskelet. Med.*, 2025.
- [31] Kuang, H., Chen, L., Huang, M., Chen, J., “Management of AIS: Etiology & rehabilitation,” *Front. Pediatr.*, 13, 1596400, 2025.
- [32] Bottino, L., Settino, M., Promenzio, L., Cannataro, M., “Deep learning for scoliosis diagnosis: Methods & databases,” *LNCS*, Springer, 2025.
- [33] Alfraihat, A., Samartzis, A., Al-Anazi, A., et al., “Curve progression prediction using Random Forest,” *PLoS One*, 17(8), e0273002, 2025.
- [34] Ohyama, S., Tsuji, T., Saito, S., et al., “Machine learning to predict future curve using visit-1 & visit-2 data,” *Eur. Spine J.*, 2025.
- [35] Zhang, L., Pei, B., Zhang, S., et al., “Deep learning scoliosis screening—enhanced approach,” *Global Spine J.*, 2025.



ST. VINCENT PALLOTTI
COLLEGE OF ENGINEERING & TECHNOLOGY, NAGPUR
AN AUTONOMOUS INSTITUTION

Supported by
tcs **TATA**
CONSULTANCY
SERVICES



THANK YOU